communications physics

Article

https://doi.org/10.1038/s42005-024-01765-9

Quantum deep generative prior with programmable quantum circuits

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Exploiting the utility of near-term quantum devices is a long-standing challenge whereas hybrid quantum machine learning emerges as a promising candidate. Here we propose a quantum-enhanced deep generative algorithm based on programmable quantum circuit-induced quantum latent codes. To validate the effectiveness of the algorithm, we conduct optical ghost imaging experiments, collecting dataset under varying physical sampling rates. Leveraging a physically enhanced loss function and a pretrained neural network, the quantum algorithm exhibits superior reconstruction performance compared to conventional algorithms. We observe the learnable quantum latent space enhances the out-of-distribution generalization capability of the pretrained model. In the context of computer vision problems, numerical results indicate that quantum latent space can increase the generation diversity of the pretrained model. Furthermore, the algorithm outperforms classical counterparts, particularly in image inpainting and colorization, by a significant margin. Our study demonstrates the utility of hybrid quantum-classical algorithms in enhancing generalization capability, highlighting the potential of near-term quantum devices in large-scale generative artificial intelligence.

Significant developments have occurred in both software and hardware for quantum computing in recent years. Quantum algorithms are widely believed to offer speedup over their classical counterparts[1]. However, these sophisticated quantum algorithms, such as Shor's[2–5] and Grover's algorithms[6–9], require subtle problem structures. Conversely, current quantum hardware faces challenges related to qubit fidelity and coherence time. Thus, the question of whether quantum algorithms confer advantages or practical utility across a broader spectrum of problems in near-term quantum devices remains unanswered[10–14].

Quantum machine learning (QML) utilizes superposition and entanglement to enhance learning and generalization capabilities, making it a promising application capable of achieving quantum advantage[15–19]. In near-term quantum devices, two typical paradigms are quantum kernel-based machine learning[20–23] and parameterized quantum circuit (PQC)-based neural networks[24]. PQC-based quantum generative adversarial networks (QGANs) can naturally learn discrete patterns better than classical GANs[25–30]. Notably, Hu et al.[31] experimentally train a quantum generator to learn the statistics of quantum data in a superconducting quantum computer, while Huang et al.[32] propose a QGAN for classical image generation. Both algorithms employ PQC as the quantum generator to learn the distributions of quantum or classical data[25,27]. However, due to current quantum hardware limitations, scaling up the width and depth of the PQC is challenging, restricting data generation tasks to low-dimensional datasets. To fully exploit near-term quantum devices for practical high-dimensional data generation, Rudolph et al.[33] experimentally demonstrate that a quantum-circuit-based generative model can learn and sample the prior distribution of a generative adversarial network to enhance image generation quality. This proposal validates the efficacy of quantum circuits in high-dimensional data generation.

The task of learning distributions generated by quantum circuits proves challenging for classical computers[34–39], potentially conferring computational advantages in specific distribution learning problems[40]. While quantum annealer-based autoencoders[41,42] and non-programmable Boson samplers[28,43] have been proposed to explore quantum advantage, the programmable capability of quantum simulators remains limited, hindering the full exploitation of generative model enhancements. However, there is a paucity of investigations into leveraging PQCs as programmable quantum priors to enhance deep generative algorithms across a spectrum of practical applications, including ghost imaging, image reconstruction and manipulation.

In this study, we propose a quantum deep generative prior (QDGP) algorithm, optimizing the use of programmable quantum priors as the latent space for data distributions, with classical neural networks handling the data generation aspect. This design makes QDGP well-suited for near-term quantum devices. We employ the pre-trained generator of BigGAN as the backbone for the deep generative model, treating PQC-generated

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distributions as the quantum latent space. Through joint optimization of the PQC and BigGAN, tailored to a problem-specific loss function, we investigate high-resolution ghost imaging reconstruction in undersampling regimes and image restoration and manipulation in common computer vision tasks.

In the ghost imaging (GI) task, we employ a physical forward model as the inductive bias to design the loss function, optimizing the generator and quantum latent space. Leveraging the classical generator, QDGP successfully recovers different resolution images without altering the model structure. Extensive experimental data demonstrates that QDGP outperforms DGP in imaging reconstruction. When recovering objects not in the pretrained data distribution, DGP generalizes poorly, whereas PQC can be trained to match the unknown target latent distribution, fully leveraging pretraining knowledge. This discovery is significant for improving the generalization of current large deep learning models. In common computer vision tasks, when assessing tasks such as image category transfer, inpainting, colorization, and super-resolution using peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) metrics, QDGP also demonstrates superior generalization capability, especially in inpainting and colorization. Therefore, the proposed quantum-enhanced latent space, combined with the pretrained large model, enhances the generalization capability of large machine learning models and can be applied to domain adaptation and transfer learning[44].

Results

Classical approaches to leveraging generative priors for image recovery and manipulation have yielded impressive outcomes. The deep image prior (DIP) demonstrates that the structural characteristics of convolutional neural networks inherently capture image statistics[45,46], effectively treating these statistics as a form of image prior. GANs are extensively employed for modeling complex data, particularly natural images[47,48]. The inherent manifold of natural images can be approximated by integrating a learnable generator G and a latent space $\mathcal{Z}$. Consequently, an image is generated by sampling a latent code $\boldsymbol{z}$ from $\mathcal{Z}$ and applying G as $G(\boldsymbol{z})$. The DIP method fixes the latent space and optimizes G based on the loss function. One limitation of DIP is that the restoration process primarily relies on existing statistics in the input image. Therefore, applying DIP to tasks requiring more general statistics is impractical.

General approach

The DGP method enhances DIP by concurrently optimizing the latent space and the generator, thereby extending the manifold of GANs. DGP employs a learnable Gaussian prior as the latent space[49]. In this context, we introduce the QDGP method, utilizing parameterized quantum circuits to generate a prior distribution as the quantum latent space to enhance GAN performance in image processing tasks. Notably, the programmable quantum latent space operates in an exponentially large quantum feature space. The PQC expectation model induced latent code can theoretically approximate any continuous multivariate distributions with an intrinsic parametric constraint[50]. The detailed description is presented in the Supplementary Note 5. Therefore, it has potential to expand the GAN manifold to the boundaries of the natural image manifold.

In near-term quantum devices, a PQC typically consists of variational quantum layers and entangling layers. Represented by $\mathbf{U}$, the functionality of multiple quantum layers is denoted. Given an initial quantum state ρ_0, the learned quantum state $\rho(\boldsymbol{\theta})$ is expressed as

$$\rho(\boldsymbol{\theta}) = \mathbf{U}(\boldsymbol{\theta})\rho_0\mathbf{U}^{\dagger}(\boldsymbol{\theta}), \tag{1}$$

where $\boldsymbol{\theta} \in \mathbb{R}^K$ denotes the quantum parameters, K is the total number of trainable quantum parameters. By choosing some observables $\mathbf{M}$ to measure the quantum state, quantum statistics can be obtained as

$$\langle \mathbf{M} \rangle = \mathrm{Tr}(\rho_{\boldsymbol{\theta}}\mathbf{M}), \tag{2}$$

where $\langle \cdot \rangle$ is the ensemble average.Viewing the expectation of $\mathbf{M}$ as the sampled latent code from the quantum Hilbert space, note that a PQC can naturally learn a Gaussian distribution by optimizing parameters, making the PQC-generated latent code more general than Gaussian latent codes. The detailed noise-free and noisy PQC structures can be found in the Methods and Supplementary Note 1. We note that PQC with physically inspired encoding, such as Heisenberg-evolution encoding, generates a quantum distribution challenging to simulate for classical computers.

Considering an original image $\mathbf{x}$ and a degraded image $\hat{\mathbf{x}} = (\mathbf{x})$ with the degradation transformation $\Phi(\cdot)$, where Φ could be a graying transform or a computational imaging process like ghost imaging, we aim to optimize the image reconstruction process. Utilizing the statistics of $\mathbf{x}$ stored in some prior, we seek the optimal $\mathbf{x}$ that best matches $\hat{\mathbf{x}}$. The GAN generator, trained on large-size natural images, serves as a strong candidate for an image prior. By integrating the quantum latent space and GAN inversion, the image reconstruction process is realized by optimizing the objective

$$\begin{aligned} \boldsymbol{\theta}^*, \boldsymbol{\xi}^* &= \arg\min_{\boldsymbol{\theta},\boldsymbol{\xi}} \mathcal{L}(\hat{\mathbf{x}}, \Phi(\mathbf{x})), \quad \mathbf{x}^* = G(\mathbf{z}^*; \boldsymbol{\theta}^*), \\ &= \arg\min_{\boldsymbol{\theta},\boldsymbol{\xi}} \mathcal{L}(\hat{\mathbf{x}}, \Phi(G(\langle \mathbf{M}(\boldsymbol{\theta}) \rangle; \boldsymbol{\xi}))), \end{aligned} \tag{3}$$

where $\mathcal{L}$ is a distance metric, G is pretrained generator parameterized by $\boldsymbol{\xi}$. Leveraging a programmable quantum prior and trainable generator significantly extends the GAN manifold, making it more flexible to fit a single image. However, fixing the generator requires fitting the entire data manifold, posing a challenging task. Fine-tuning the generator may lead to the deterioration of generated images. Preserving the generative prior hinges on a refined distance metric and optimization strategy.

Quantum deep generative prior for ghost imaging

GI is a computational imaging technique enabling image acquisition under low-light conditions through single-pixel measurements. GI holds significant promise for imaging in photon-limited scenarios and spectral regions where array cameras are impractical. Traditional imaging reconstruction methods, such as differential GI, involve calculating the correlation between bucket measurements and corresponding modulation patterns to reconstruct images[51–55]. Compressive sensing methods have also been proposed to reduce sampling requirements by leveraging sparse constraints on the unknown object[56,57]. In parallel, supervised machine learning approaches[58,59] aim to enhance imaging quality under low sampling regimes. However, these machine learning methods often demand large-scale datasets, a limitation in practical scenarios.

To address the challenges of dataset requirements and improve imaging quality, we employ a large-scale dataset pretrained GAN as the inversion process for image recovery. Additionally, we integrate the physical forward model as an inductive bias to compute the loss function, jointly fine-tuning the pretrained GAN and the quantum programmable prior. The schematic of the proposed algorithm is illustrated in Fig. 1. This approach simultaneously leverages the knowledge of the pretrained model and the physical forward model, placing it within the realm of data-physics hybrid-driven methods[60]. Formally, in GI, the imaging degradation process $\Phi(\cdot)$ is represented by a linear transformation $\mathbf{H}$ corresponding to a set of randomly modulated light field speckles. The degraded image $\hat{\mathbf{x}}$ is expressed as a collection of experimental bucket signals $\mathbf{B}$. Consequently, Eq. (3) can be redefined as

$$\boldsymbol{\theta}^*, \boldsymbol{\xi}^* = \arg\min_{\boldsymbol{\theta},\boldsymbol{\xi}} \left\| \mathbf{B} - \mathbf{H} \cdot G(\langle \mathbf{M}(\boldsymbol{\theta}) \rangle; \boldsymbol{\xi}) \right\|^2, \tag{4}$$

where we choose the L_2-norm as the distant metric. In practice, we generally add an extra constraint to Eq. (4) to enhance the prior information. Total variation (TV) constraint can render GAN recovered image more smooth since images meet local smooth characteristic. Eq. (4) thus can be further

written as

$$\boldsymbol{\theta}^*, \boldsymbol{\xi}^* = \arg\min_{\boldsymbol{\theta},\boldsymbol{\xi}} \|\mathbf{B} - \mathbf{H}\tilde{\mathbf{x}}\|^2 + \lambda\|\tilde{\mathbf{x}}\|_{\mathrm{TV}}, \tag{5}$$

where we define GAN reconstructed image as $\tilde{\mathbf{x}} = G(\langle O(\boldsymbol{\theta})\rangle; \boldsymbol{\xi})$, λ is the coefficient hyperparameter. Upon convergence of the optimization process, we obtain $\tilde{\mathbf{x}} = \mathbf{x}^* = G(\langle O(\boldsymbol{\theta})\rangle; \boldsymbol{\xi})$. The linear transformation $\mathbf{H}$ encodes the physical information of the GI system. The pretrained GAN G and the quantum latent space encapsulate the image prior derived from large-scale datasets. Therefore, the optimization equation Eq. (5) integrates the inductive bias of physics and datasets, holding significant potential for reducing sampling requirements and enhancing generalization[61].

To assess the efficacy of QDGP with a physically informed loss function, we conduct initial optical experiments, capturing bucket signals under varying sampling rates. Biological samples were selected as target images, deliberately chosen outside the pretrained datasets, without correlated samples or categories. The exploration of image priors based on the physical forward model in this context has not been previously undertaken, posing a crucial challenge in practical scenarios, particularly in microscopic contexts. Leveraging the flexibility of GAN, we successfully reconstruct object images at different resolutions by applying distinct downsampling factors to GAN-generated images (see Methods). The experimental setup of ghost imaging can be found in Supplementary Note 7.

In our first set of experiments, we focus on the sample "butterfly" to evaluate the performances of QDGP, DGP, and DGI, utilizing the BigGAN generator with 256^2 resolution as the neural network backbone[62]. As shown in Fig. 2, QDGP consistently outperforms DGP and DGI across common metrics such as Peak signal-to-noise ration (PSNR) and structural similarity (SSIM). Here, PSNR is defined as

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right), \tag{6}$$

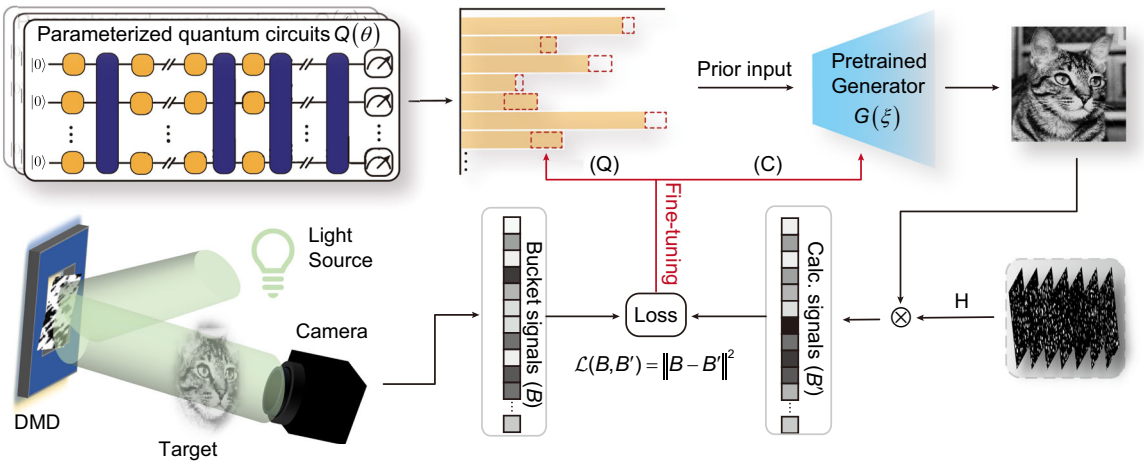


Fig. 1 | Quantum deep generative prior for ghost imaging. The parameterized quantum circuit induces the quantum latent space and is followed by a pretrained generative adversarial network to recover the images. The recovered image then is mapped to the calculated bucket signals by using a prescribed random illumination pattern i.e. the physical model. By calculating the loss between the experimental and calculated bucket signals, the minimization can be achieved by using the gradient descent method.

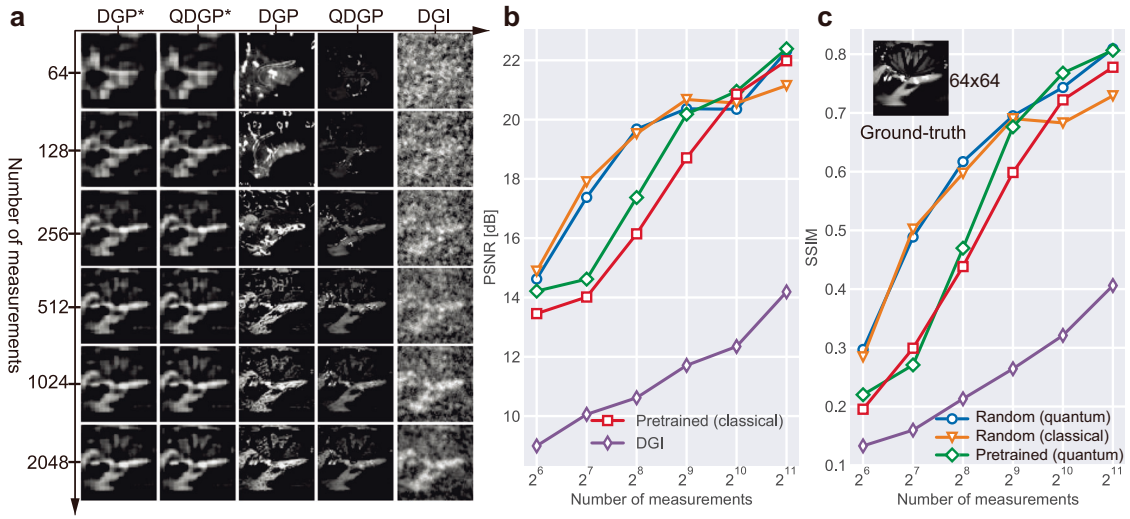


Fig. 2 | The experimental results of different image reconstruction methods. DGP (deep generative prior), QDGP (quantum deep generative prior) and DGI (differential ghost imaging). **a** DGP* and QDGP* denote that GAN generator is random initialized. In contrast, DGP and QDGP adopt the pretrained generator of BigGAN. **b** The metrics of peank signal-to-signal ratio (PSNR) and structural similarity (SSIM). **c** calculated by the reconstructed image and the ground-truth image.

where MSE is the mean square error between the noise-free image I and noisy image K with scale $m \times n$ and it can be written as

$$\mathrm{MSE} = \frac{1}{mn}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}\left[I(i,j) - K(i,j)\right]^2. \tag{7}$$

MAX_I is the maximum possible pixel value of the image. When the image are presented using pulse-code modulation with B bits, $\mathrm{MAX}_I = 2^B - 1$. Besides, SSIM can be defined as

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}, \tag{8}$$

where x, y are two images, μ_x, μ_y are the mean value of two images, σ_x^2, σ_y^2 are the variance of x, y, $\sigma_{x,y}$ is the covariance of x and y, $c_1 = (k_1\mathrm{MAX}_I)^2$, $c_2 = (k_2\mathrm{MAX}_I)^2$ are two variables to stabilize the division with weak denominator. Generally, we have $k_1 = 0.01$, $k_2 = 0.03$.

In scenarios with a small number of measurements, the randomly initialized GAN generator with classical and quantum priors exhibit competitive performance. It is worth noting that the randomly initialized GAN generator, with its large number of parameters, faces challenges in recovering high-quality images based on a small number of measurements, i.e., a low-dimensional loss constraint. However, with an increase in the number of measurements to 2^9 (sampling rate of 12.5%), both QDGP and DGP produce higher-quality images, and QDGP demonstrated superior metrics in PSNR and SSIM (see Fig. 2b). For the pretrained GAN generator, both DGP and QDGP perform less effectively at small sample rates. We speculate that, given the sample's absence in the pretrained datasets and the random specification of the label, the GAN generator encounters difficulties in matching the correct image pattern with a low-dimensional loss constraint. Pretrained generators require more information to fine-tune the parameters, resulting in inferior performance compared to randomly initialized GANs. Intriguingly, QDGP outperforms DGP, showcasing that PQC has a more flexible distribution generation ability and can achieve better-optimized images toward the intersection of the natural image manifold and the zero-distance manifold. As the sampling rate increases, QDGP maintains larger SSIM and PSNR values than DGP. Additionally, pretrained models reconstructs higher-quality images than randomly initialized ones. Due to the higher-dimensional loss constraint, the optimization process more effectively transcends the boundaries of the GAN manifold and moved towards the manifold of ground-truth targets. In contrast, randomly initialized models may not initiate the optimization process from the natural image manifold. Although they still approach the ground-truth target, the recovered image exhibits a slightly more mosaic appearance. Furthermore, both DGP and QDGP methods outperform DGI, highlighting the GAN network's inherent advantage in handling images.

To evaluate QDGP's performance on larger resolution images, we conduct optical GI experiments to capture bucket signals, choosing to reconstruct images of a biological sample, the wasp's wing. It's important to note that this image lies outside the natural image manifold, characterized by numerous textures and details. The target image has dimensions 128×128, considers high resolution for GI systems, necessitating a prolonged time for acquiring bucket signals. We investigate samplings with $M \in \{256, 512, 1024, 2048\}$ to assess the performance of randomly initialized and pretrained QDGP. Experiment results are presented in Fig. 3.

In high-resolution reconstruction of the biological sample, pretrained models generally perform worse than their randomly initialized counterparts, particularly evident in the SSIM metric (Fig. 3c). As shown in Fig. 3a, pretrained DGP produces a blurred image, while QDGP reconstructs a more refined wing object. In the regime of a large number of measurements ($M > 2^{10}$), DGP even performs worse than the DGI method in terms of the SSIM metric. QDGP maintains the advantage of machine learning methods over DGI across all measurement numbers. The potential reason is the significant difference between biological samples and natural images. Although pretrained models store considerable prior information about natural images, generalizing well to unseen biological images using only a Gaussian latent space is challenging. However, we observe that PQC-induced quantum latent space can alleviate this challenge and improve out-of-distribution generalization ability for unknown biological samples.

In terms of the PSNR metric (Fig. 3b), both pretrained QDGP and DGP outperform their randomly initialized counterparts and the conventional DGI method. The result is surprising given the opposite result obtained with the SSIM metric. It is essential to note that SSIM evaluates the overall similarity of two images, while PSNR evaluates the MSE. In cases where the noise level of the imaging system is high, the reconstructed image by DGI is noisy (Fig. 3a). However, machine learning can leverage the GAN image prior for denoising[63], resulting in a cleaner image. Additionally, pretrained and randomly initialized QDGP achieve a higher PSNR value than DGP with pretrained and randomly initialized GAN. When using a pretrained model, the performance gap between QDGP and DGP is amplified compared to the case of a randomly initialized model. We present additional results of QDGP on large number of MNIST digits (see Supplementary Note 2). The results consistently demonstrate the potential advantage of QDGP in enhancing the generalization ability of pretrained models. In Supplementary Note 2, we also present the results of noisy PQC-

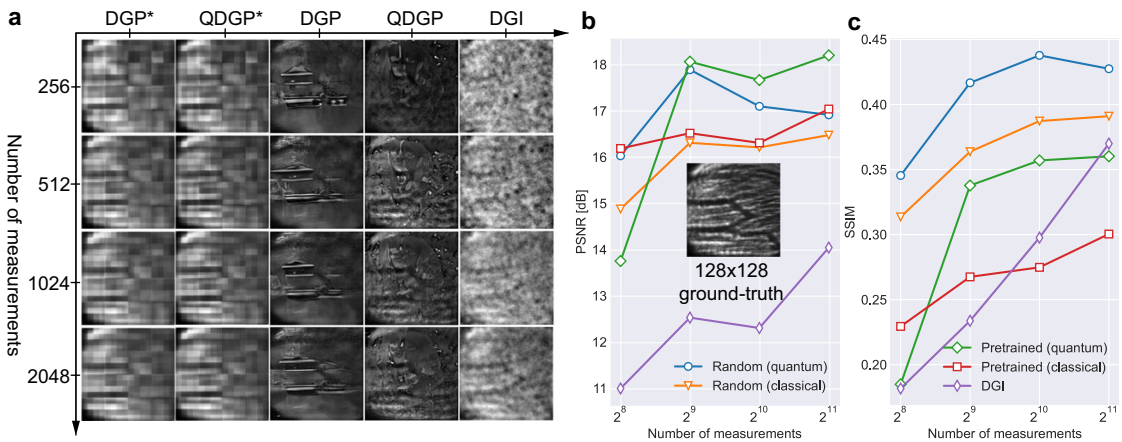


Fig. 3 | The reconstruction performance of large resolution biological sample with varied number of measurements based on different methods. a The reconstructed images with resolution 128×128 by random initialized or pretrained QDGP and DGP and DGI method, where * denotes the random initialized generative adversarial network generator. **b** The peak signal-to-noise ratio (PSNR) metric of the reconstructed image and the ground-truth image for different methods. **c** The structural similarity metric (SSIM) of the reconstructed image and ground-truth image for different methods.

enhanced QDGP to analyze robustness. As expected, quantum noise degrades the performance of QDGP, especially for the pretrained generator.

In Supplementary Note 3-4, we examine the outcomes of Matrix Product State (MPS)-based PQC with a low bond dimension (16) and state-vector-based PQC for deep circuit layers. We observe a smooth decrease in loss as the iterations increase when the layer count increases from 6 to 10. These findings suggest that QDGP does not encounter the Barren Plateau in our experimental setup. We opted not to further escalate the layer count, as it led to a reduction in the PSNR of the reconstructed image from state-vector-based PQC. Interestingly, the PSNR of MPS-based PQC remains relatively stable across different layers. This indicates that MPS with a low bond dimension can only approximate highly entangled states. Given that excessive entanglement (over-expressive PQC) can trigger the Barren Plateau, we deduce that MPS-based PQC with a low bond dimension may restrict the expressiveness of PQCs, thereby circumventing the Barren Plateau phenomenon. However, state-vector-based PQC with around 4 or 6 layers exhibits a higher PSNR than MPS-based PQC. Thus, MPS can serve as a diagnostic tool to assess the risk of PQC encountering the Barren Plateau and aid in selecting suitable layers for variational quantum circuits.

We visualize the density of the latent space during optimization, observing the start and end iterations. From Fig. 4a, b, we observe that the sampled density resembled a Gaussian distribution, with different distributions at the start and end iterations, concentrated mostly in the range [−0.5, 0.5]. In contrast, the sampled density of the quantum latent space is more decentralized for both pretrained and randomly initialized models (Fig. 4c, d). These results suggest that quantum latent space is more flexible than Gaussian latent space, likely contributing to the enhancement of out-of-distribution generalization for PQC-generated distributions. Moreover, the density variation between the start and end iterations of QDGP is larger than that of DGP. For samples not in the pretrained dataset, this significant variation can aid in finding the optimization path to achieve optimal reconstruction.

Quantum deep generative prior for imaging restoration and manipulation

To assess the performance of QDGP in imaging restoration and manipulation, we select four standard tasks: category transfer, super resolution, colorization, and inpainting. The schematic of QDGP for image restoration and manipulation is depicted in Fig. 5. Detailed algorithm information is available in the Methods section.

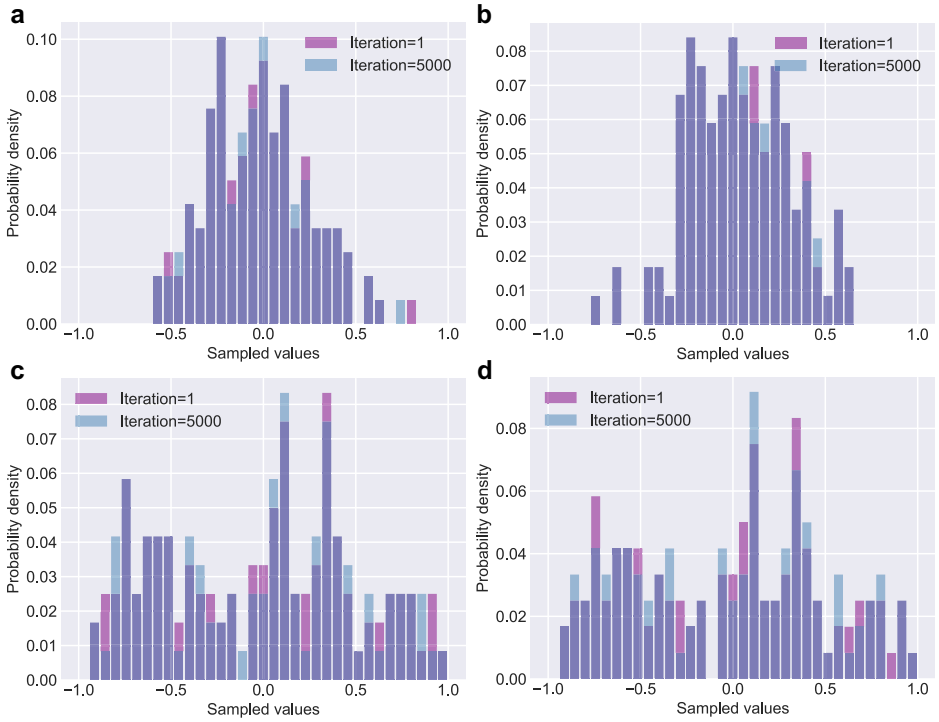


Fig. 4 | The probability density of deep generative prior and quantum deep generative prior with random and pretrained initialization. a The density of classical latent space for random initialized generator. **b** The density of classical latent space for pretrained generator. **c** The density of quantum latent space for random initialized generator. **d** The density of quantum latent space for pretrained generator.

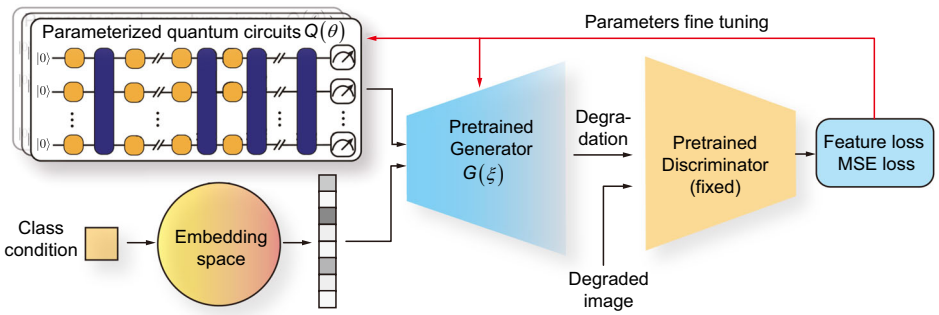


Fig. 5 | Parameterized quantum circuit coded latent space for deep generative prior with pretrained generator and discriminator. The feature loss and mean square error loss are both calculated to obtain a weighted loss. Subsequently, the quantum and classical parameters of parameterized quantum circuit and generator are fine tuned to minimize the loss according to gradient descent methods.

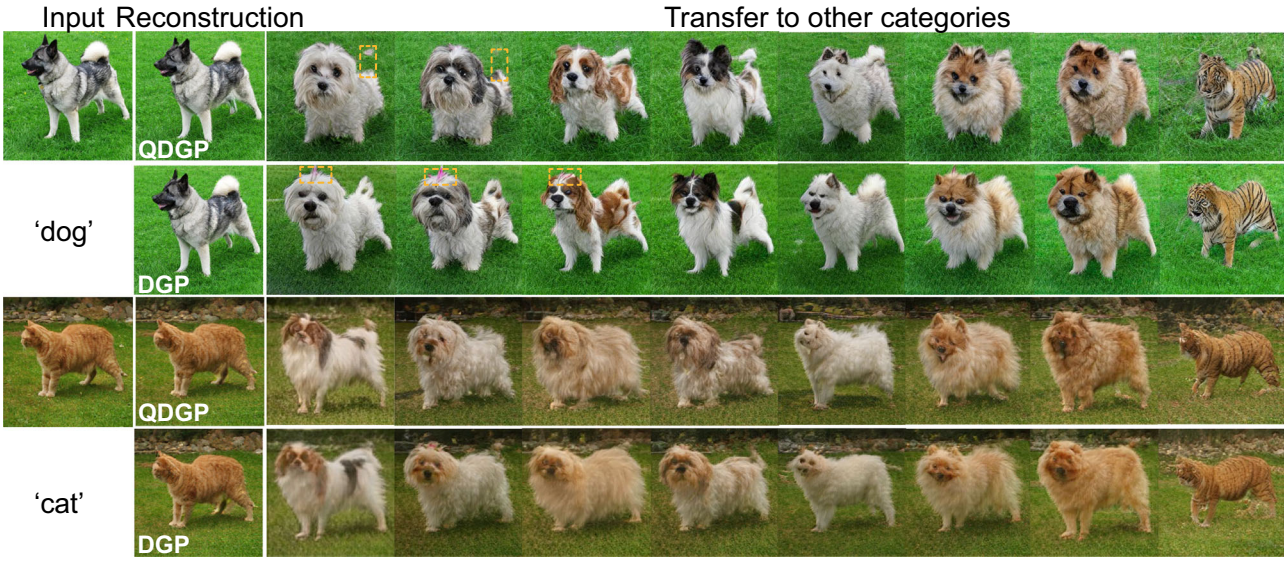


Fig. 6 | The performance of category transfer with quantum deep generative prior (QDGP) and deep generative prior (DGP) methods. QDGP and DGP enable the editing of semantics of objects in images by altering the input class condition. Quantum latent space code has different category transfer performance with Gaussian latent space. The yellow dashed boxes mark the imperfect image details.

Category transfer. In BigGAN, the class condition governs the generated content in the pre-trained neural network through the embedding layer. Initially, a predefined class condition is input into the GAN neural network, and subsequently, the loss between the reconstructed image and the target image for transfer is minimized. After fine-tuning the latent code and generator parameters, the network can generate content from other categories when provided with different class conditions. In Fig. 6, category transfer experiments are conducted, transferring the object category of given images. Both QDGP and DGP successfully transferred 'dogs' and 'cats' to various other categories while preserving pose, size, and image configurations. QDGP exhibits competitive performance with DGP, maintaining similarity in background, pose, and size to the input image. However, QDGP and DGP show differences in transfer details, particularly in facial and hair features. It is important to note that direct comparisons of superiority cannot be made, as the evaluation of category transfer effectiveness relies on visual inspection. Both methods demonstrate impressive performance, with yellow dashed boxes highlighting the reconstruction differences between QDGP and DGP.

Inpainting. The image inpainting task aims to restore the contaminated area of an image using neighboring information. Generally, the degradation transformation can be expressed as the Hadamard product of the original image and the binary mask $\mathbf{m}$: $\Phi(\mathbf{x}) = \mathbf{x} \odot \mathbf{m}$, where $\odot$ denotes Hadamard product. Figure 7 illustrates the recovery of missing portions in harmony with the overall image context, leveraging the generated image prior. It is worth noting that, typically, DGP performs better for images with periodic patterns, where neural networks can capture regular spatial information for improved recovery[49]. However, in scenarios without periodic patterns, we assessed the performance under random class and designated class conditions. Contrary to expectations, designating a class condition did not consistently yield superior performance, as depicted in Fig. 7. In the case of inpainting 'llama' and 'dog', QDGP without a class condition (QDGP(-1)) exhibited higher PSNR and SSIM values than QDGP with a designated class condition. The inpainting performance is influenced by the randomly generated image and the location of the missing pixels. Quantum latent code, possessing distinct distribution features from Gaussian code, inherently generates images different from DGP, thus increasing the diversity of the generated images. In the early stage, the inpainted image exhibits lower quality, but through optimization using MSE loss, the abnormal part is refined to match the input image. In contrast, Gaussian code ensures generated images lie within the natural image manifold but constrains diversity. From this perspective, QDGP can decouple the relationship between inherent patterns and latent code, enhancing diversity to recover missing parts more effectively. Additional experiments were conducted on the ImageNet validation set using QDGP with a designated class condition for performance evaluation. In Supplementary Note 2, we present the reconstruction process of image inpainting of QDGP algorithm under the configuration of random and predefined category labels.

Colorization. Image colorization aims to restore a grayscale image $\hat{\mathbf{x}} \in \mathbb{R}^{H\times W}$ to a colorful image with RGB channels $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$. The degradation process Φ preserves only the brightness of the original image. By incorporating the degradation transform into Eq. (3), the objective is to find the colorful image $\mathbf{x}^*$ with the same grayscale representation as $\hat{\mathbf{x}}$. Through fine-tuning the GAN generator and the quantum latent space to minimize the progressive discriminator loss, Eq. (3) can be efficiently solved to restore the colorful image. It is essential to note that, as the grayscale information is accurate, the task involves restoring the color information, specifically the 'ab' dimension of the image. The final colorful image is synthesized by combining the true gray information with the learned color information. Experimental results are presented in Fig. 8. The results indicate that QDGP outperforms DGP in terms of PSNR and SSIM metrics in the colorization task. QDGP is more likely to exhibit improved color and style consistency compared to DGP. Notably, in the colorization of 'bird' and 'boat' images (i.e., the second and third rows), QDGP demonstrates more refined color consistency, conveying a scene reminiscent of autumn. In contrast, DGP tends to conform to the color of the ground-truth, limiting its ability to generate diverse images of different types and styles compared to QDGP. QDGP showcases a stronger semantic generating ability and achieves a more natural colorization in image details, as visualized in the dashed white boxes of Fig. 8.

Super-resolution. Image super-resolution aims to generate a high-resolution image $\mathbf{x} \in \mathbb{R}^{3\times kH\times kW}$ given a low-resolution image $\hat{\mathbf{x}} \in \mathbb{R}^{3\times H\times W}$, where k denotes the upsampling factor. In this task, we



Fig. 7 | Inpainting performance of quantum deep generative prior (QDGP) and deep generative prior (DGP) with or without class condition information. QDGP(-1) and DGP(-1) are the cases with random class input. QDGP shows a better inpainting performance in non-ImageNet images from the first row to last row. Without class information, QDGP even performs better in the third and fourth row, showing a strong learning capability.

initially apply a downsampling process to execute the degradation transform and then optimize Eq. (3) to generate the high-quality image. The super-resolution in this context is accomplished by leveraging the generative prior of pretrained knowledge, a distinction from the physical super-resolution observed in optical imaging systems.

In Fig. 9, we showcase four super-resolved images produced by QDGP and DGP. The input class condition corresponds to the image category. DGP and QDGP exhibit competitive performance in the super-resolution task, with highly similar visual effects. Evaluation based on image metrics such as PSNR and SSIM indicates that QDGP slightly outperforms DGP in the presented four images. For instance, in the last row of Fig. 9, QDGP demonstrates superior super-resolution performance on the bee's wing compared to DGP.

The performance of different tasks of QDGP and DGP on the 1k ImageNet validation set is presented in the Supplementary Note 2 (see Supplementary Tables I, II and III). The impact of quantum noise on the final performance of QDGP in image restoration and manipulation tasks are analyzed in Supplementary Note 2. We adopt a practical noise rate and depolarization quantum channel to characterize the noisy quantum computation. We find the practical quantum noise has no large negative impact on the final performance. For given four tasks, noisy QDGP all showcases the impressive performance. The result is quite intuitive since the quantum latent space is not directly responsible for the image generation. The role of quantum latent space is to increase the diversity and complexity of the latent space. Quantum noise can reduce the expressive power and for large noise rate, it will be degraded to uniform distribution. In our setting, the complexity and diversity of noisy PQC generated latent space are still maintained to obtain high quality restoration.

Conclusions

In summary, we propose a quantum latent code enhanced deep generative algorithm called QDGP based on PQCs. Given the intrinsic difficulty in learning distributions generated by PQC, the quantum latent space exhibits the potential to enhance the expressive power and out-of-distribution generalization of deep generative models. To assess QDGP's performance in practical applications, we initially investigate its application in GI under an undersampling regime. We propose a loss function induced by a physics-forward model and utilize a pretrained BigGAN generator. By leveraging both data and physical prior information, QDGP-enhanced GI effectively reconstructs high-quality non-natural images based on a pretrained model trained on natural images. In contrast, the DGP method struggles to reconstruct high-quality non-natural images. Furthermore, QDGP with a randomly initialized generator outperforms DGP with a randomly initialized generator, indicating that even without using data knowledge, QDGP still maintains an advantage over DGP. These results demonstrate that the quantum latent code enhances the generalization capability of large pretrained models. In addition, QDGP-enhanced GI maximally utilizes prior information from data, the generative model, and physics to flexibly reconstruct high-quality images for arbitrary objects under an undersampling regime. QDGP with the physical forward model $\mathbf{H}$ as the inductive bias can make fully use of the physics information to recover the images. Since using GI, we can simply regard $\mathbf{H}$ as the imaging process and all the information involving the spatial freedom are ignored. However, our method can also be applied to cases where the forward model is known but the inverse process is unknown and difficult to solve, as seen in refs. 64,65, where the forward model relies on refined electromagnetic modeling. We can use quantum machine learning to pre-parameterize the inverse process



Fig. 8 | Colorization performance of quantum deep generative prior (QDGP) and deep generative prior (DGP) for various images. The first and second row are the non-Imagenet images. Dashed white boxes highlight the differences between QDGP and DGP. White dashed boxes highlight the reconstruction difference.

and employ the physical forward model as the inductive bias to update the parameters. From this perspective, our proposed method has numerous applications in solving inverse problems.

Moreover, we conduct extensive experiments with QDGP on computer vision problems such as category transfer, colorization, inpainting, and super-resolution. Interestingly, we observe that in colorization and inpainting problems, QDGP outperforms DGP by a significant margin. In the other two tasks, QDGP exhibits competitive performance against DGP. We postulate that the quantum latent code increases the generation diversity of pretrained models by decoupling the inherent relationship between the latent code and the generative model. Under the constraints of feature and MSE loss, the final recovered image aligns more closely with the ground-truth image. In contrast, DGP with a Gaussian latent code operates within a limited pattern space. The distribution generated by PQC has more freedom and can also be trained to fit the Gaussian code[66]. From this perspective, QDGP surpasses its classical counterparts, achieving state-of-the-art performance.

QDGP integrates a pretrained large-scale machine learning model with a programmable quantum circuit, making it applicable to hybrid quantum-classical high-performance computing clouds[67]. In future work, we will focus on studying quantum-enhanced generative artificial intelligence such as generalization[68,69] to explore the practical advantages of quantum computation.

Methods

Parameterized quantum circuits

Typically, a parameterized quantum circuit consists of a quantum encoding layer $\mathbf{E}$ and a quantum learning layer $\mathbf{V}$. In this work, we represent the joint operation of the encoding and learning layers as $\mathbf{U}$, where $\mathbf{U}(\boldsymbol{\theta}) = \mathbf{V}(\boldsymbol{\theta})\mathbf{E}$. It's important to note that the quantum encoding layer does not possess parameters. Our objective is to map a uniformly sampled vector to a quantum latent code through a PQC. The required number of qubits for amplitude encoding depends on the quantum encoding strategy and is given by $\lceil \log_2 d \rceil$, where d is the dimension of the latent code. While amplitude encoding reduces the number of qubits exponentially, its practical implementation faces challenges such as the need for quantum random access memory and the requirement for exponential shots for quantum state tomography. A more practical approach is angle encoding, using one qubit

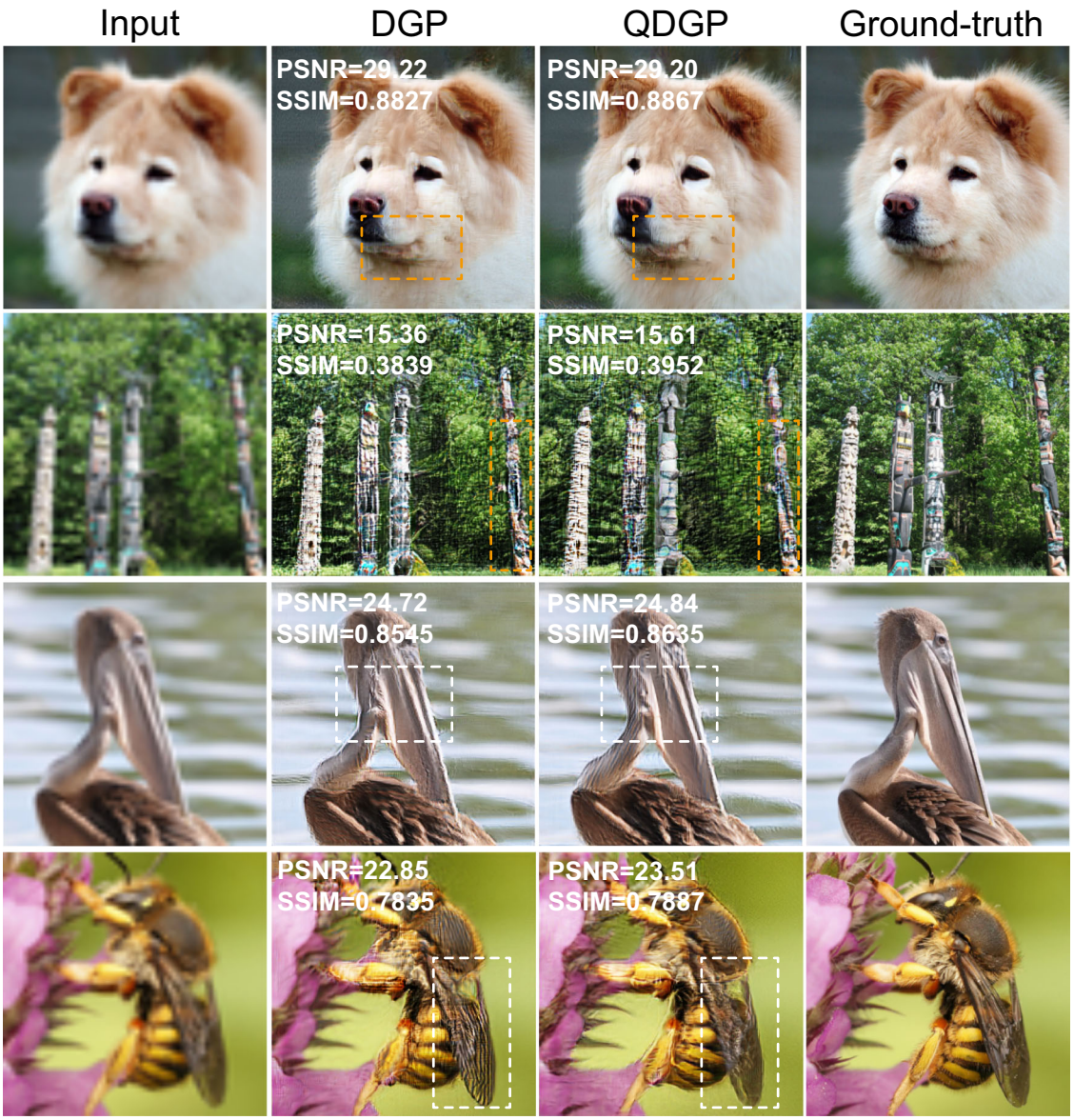


Fig. 9 | Superresolution performance comparison of quantum deep generative prior and deep generative prior in various images. Note that the third and fourth row are the images not in the Imagenet. Input image is a low resolution with size of 64 × 64. The super-resolved images are of size 256 × 256. Yellow and white dashed boxes highlight the superresolution difference.

to encode one element. We employ the data-reuploading encoding style to enhance the expressive power of PQC. For a uniformly sampled vector $\mathbf{s} \in \mathbb{R}^d$, the quantum encoded state is given by:

$$\begin{aligned} |\mathbf{s}\rangle &= \bigotimes_{i=1}^{N} \mathbf{R}_x(\arcsin(\mathbf{s}_i))\mathbf{R}_y(\arcsin(\mathbf{s}_i))|0^N\rangle \\ &= \mathbf{E}(\mathbf{s})|0^N\rangle. \end{aligned} \tag{9}$$

Here, $N = d$ and $|0^N\rangle = |0\rangle^{\otimes N}$. The quantum learning operator of the l-th layer is defined as

$$\mathbf{V}(\boldsymbol{\theta}_l) = \bigotimes_{i=1}^{N-1} \mathbf{CZ}_{i,i+1} \bigotimes_{i=1}^{N} \left[\mathbf{R}_z\left(\theta_l^{i,1}\right)\mathbf{R}_y\left(\theta_l^{i,2}\right)\mathbf{R}_z\left(\theta_l^{i,3}\right)\right], \tag{10}$$

where $\boldsymbol{\theta}_l = \{(\theta_l^{i,1}, \theta_l^{i,2}, \theta_l^{i,3})\}_{i=1}^{N}$, and $\theta \in [0, 2\pi]$ denotes the quantum parameters of PQC in the l-th learning layer. **CZ** is two-qubit gate, which can render two qubits entangled. One can also choose **CNOT** gate, exponential **ZZ**, **XX** gates. **ZZ** and **XX** gates can be written as the combination of **CNOT** gate and rotation Pauli gate. Detailed description can be found in Supplementary Note 1. We adopt a hardware-efficient circuit structure to construct the quantum circuit for increased feasibility and ease of deployment on NISQ devices.

The final quantum state with data-reuploading encoding style is given by:

$$|\psi(\mathbf{s}, \boldsymbol{\theta})\rangle = \mathbf{V}(\boldsymbol{\theta}_L)\prod_{l=1}^{L-1}\mathbf{E}(\mathbf{s})\mathbf{V}(\boldsymbol{\theta}_l)|0^N\rangle, \tag{11}$$

where L is the number of trainable layers of the PQC. The quantum latent code is obtained by measuring $|\psi\rangle$ to obtain the quantum statistics. Using a Hermitian observable $\mathbf{M}$, we estimate the expectation value using quantum expectation estimation method:

$$\langle \mathbf{M} \rangle_{\mathbf{s},\boldsymbol{\theta}} = \sum_j w_j h_j(\mathbf{s}, \boldsymbol{\theta}), \tag{12}$$

where we define $\mathbf{M} = \sum_j w_j h_j$ with $h_j \in \mathcal{P}_N$ denoting the sub-Hamiltonian in the N-qubit Pauli group $\mathcal{P}_N$. The observable expectations are considered the

quantum latent code and are then fed into the generator. For simplicity, we generally choose $w_j = 1$ and $h_j = \mathbf{Z}$ as the measurement setting, where $\mathbf{Z}$ denotes the Pauli-Z gate. It is important to note that in the main text, we use a more general density matrix description. When the initial state is pure, the density matrix formulation remains applicable.

In the case of BigGAN with 256^2 resolution, the latent code $\mathbf{z} \in \mathbb{R}^{120}$. Simulating a 120-qubit circuit is highly expensive for classical computers with state vector simulator. However, MPS and its evolution can approximate the computation of quantum circuits by using appropriate bond dimension. In the supplementary Note 3-4, we analyze the performance of QDGP-MPS in GI reconstruction. Considering BigGAN's hierarchical structure, where the latent code is split into 7 pieces, we adopt a similar circuit cutting strategy. Thus, the quantum circuit is split into 7 sub-circuits, each generating a portion of the latent code. Each circuit has 18 qubits, and additional codes are discarded. The circuit cutting strategy is inspired by the code splitting in BigGAN. When large-scale quantum computers become available, generating the high-dimensional quantum code directly as the input of the generator will further amplify quantum utility.

Neural network architectures

We employ the BigGAN neural network with 256^2 resolutions as the foundational framework for our model. BigGAN is chosen due to its remarkable performance in generating images across a variety of object categories. The architectural representation of BigGAN is illustrated in Fig. 10. Specifically, for the 256^2 resolution, the channel multiplier and batch size are configured to 96 and 1920, respectively, during the training process. Notably, we opt not to train BigGAN from the ground up; instead, we utilize a pre-trained PyTorch model[49]. The pre-training is accomplished using the ImageNet dataset.

The inclusion of a class condition is imperative as an input for the BigGAN generator. In the case of the GI task, where access to the genuine class label is unavailable, the class condition is arbitrarily set to a random embedding hidden vector $\mathbf{y}$. It's important to note that this embedding may not accurately correspond to the image intended for reconstruction. For classical image restoration and manipulation tasks, where the correct label is typically available, our focus here is on the generation task. The class label can be predicted through the training of a deep classification neural network. Consequently, the embedded hidden vector contains the correct prior information obtained from the pre-trained embedding. To enhance performance, we randomly initialize 500 latent codes $\mathbf{z}$ and assess their respective loss values. The latent code corresponding to the minimum loss value is selected as the initial latent code for the generator. For QDGP, we randomly initialize 500 PQCs and subsequently choose one with the smallest loss value[49]. It is crucial to highlight that the loss function selected is the discriminator feature loss.

Learning process

The learning process comprises two stages: optimizing the PQC $Q(\boldsymbol{\theta})$ and fine-tuning the generator $G(\boldsymbol{\xi})$. In practice, the gradients of $G(\boldsymbol{\xi})$ parameters can be updated using automatic differentiation (AD) since the operators in the generator are fully differentiable. However, the gradients of the quantum parameters of PQC cannot be calculated using AD. Generally, we adopt the parameter-shift rule to calculate gradients for each parameter[70–72]. The computational overhead for calculating gradients of quantum parameters is $O(KS)$, where K is the total number of quantum parameters, and S is the required shots of the quantum circuit to estimate the observable expectation. We employ the Pauli-Z observable $\mathbf{z} = (\langle \mathbf{Z}_1 \rangle, \cdots, \langle \mathbf{Z}_i \rangle, \cdots, \langle \mathbf{Z}_N \rangle)$ as the quantum latent code, with i denoting the ith qubit. The gradient of the loss

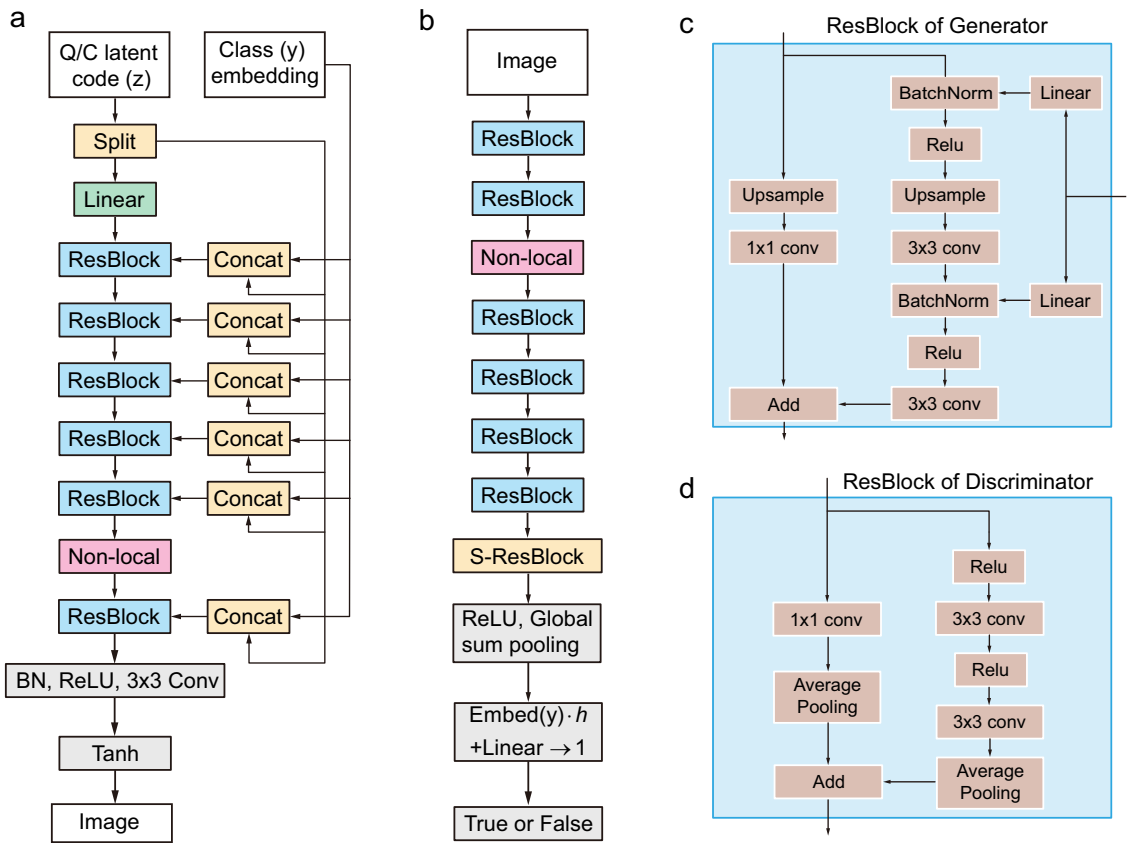


Fig. 10 | The architecture of BigGAN, including detailed structures of the generator and discriminator. a Presents a typical architecture layout for the BigGAN generator. The output generated image can have different resolutions by using extra upsampling or downsampling modules. **b** Depicts an architecture layout for the BigGAN discriminator; S-ResBlock denotes the standard residual block as per ref. 77. **c** Represents a Residual Block in the BigGAN generator. **d** Depicts a Residual Block in the BigGAN discriminator. The more detailed structure description can refer to ref. 62.

function with respect to the quantum parameters can be calculated using the chain rule

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}} = \frac{\partial \mathcal{L}}{\partial \langle \mathbf{M} \rangle} \cdot \frac{\partial \langle \mathbf{M} \rangle}{\partial \boldsymbol{\theta}}, \tag{13}$$

Note that $\partial \mathcal{L}/\partial \langle \mathbf{M} \rangle$ can be efficiently calculated by AD. $\partial \langle \mathbf{M} \rangle / \partial \boldsymbol{\theta}$ can be calculated by

$$\begin{aligned} \frac{\partial \langle \mathbf{M} \rangle}{\partial \theta_k} &= \sum_j w_j \langle 0^N | \partial_{\theta_k} \mathbf{U}^\dagger(\mathbf{s}, \theta_k) h_j \mathbf{U}(\mathbf{s}, \theta_k) | 0^N \rangle \\ &= \sum_j c w_i \left[\langle 0^N | \mathcal{M}(\theta_k + \delta) | 0^N \rangle - \langle 0^N | \mathcal{M}(\theta_k - \delta) | 0^N \rangle \right], \end{aligned} \tag{14}$$

where we define $\mathcal{M}(\mathbf{s}, h_i, \boldsymbol{\theta}) = \mathbf{U}^\dagger(\mathbf{s}, \boldsymbol{\theta}) h_i \mathbf{U}(\mathbf{s}, \boldsymbol{\theta})$. δ, c are constant coefficient independent of θ_k. Since PQC is composed of Pauli operations, we assume that the PQC is composed of $\mathbf{U}(\theta_k) = \exp(-i\theta_k \mathbf{P}/2)$. The gradient $\partial_{\theta_k} \mathbf{U}(\theta_k) = -i/2 \mathbf{U}(\theta_k) \mathbf{P}$ and then inserting into Eq. (14), we have

$$\frac{\partial \langle \mathbf{M} \rangle}{\partial \theta_k} = -\frac{i}{2} \sum_j w_j \langle 0^N | \mathbf{U}^\dagger(\theta_k) [\mathbf{P}, h_j] \mathbf{U}(\theta_k) | 0^N \rangle. \tag{15}$$

With commutation rule, i.e. $[\mathbf{P}, h_j] = i\mathbf{U}^\dagger(\frac{-\pi}{2}) h_j \mathbf{U}(\frac{\pi}{2}) - \mathbf{U}^\dagger(\frac{-\pi}{2}) h_j \mathbf{U}(\frac{-\pi}{2})$, we can derive the analytical gradient as

$$\frac{\partial \langle \mathbf{M} \rangle}{\partial \theta_k} = \sum_j \frac{w_j}{2} \left[\langle 0^N | \mathcal{M}(\theta_k^+) | 0^N \rangle - \langle 0^N | \mathcal{M}(\theta_k^-) | 0^N \rangle \right], \tag{16}$$

where $\theta_k^\pm = \theta_k \pm \pi/2$. Due to the distinctive properties of PQC, we can derive the analytical gradient of the latent code concerning quantum parameters. Recently, a more structured quantum circuit has introduced a backpropagation scaling, resulting in a reduced gradient overhead[73], which is highly advantageous for NISQ devices. In practical implementation, the parameter-shift rule can be employed to manually calculate the quantum gradient for each parameter. Subsequently, the Adam optimizer can be used to update the parameters[74].

Software and hyperparameters

The pretrained BigGAN model operates at a resolution of 256^2. The dimensionality of the latent code is 120, and the embedding dimension is 128. The model is constructed using the PyTorch package[75]. Quantum circuit simulations are conducted with the Tensorcircuit software[76], which supports vmap and just-in-time compilation. The simulation of hardware-efficient quantum circuits can accommodate a maximum of over 20 qubits. The simulations are performed on the high-performance computing server, QSIP, equipped with 192 Intel CPU cores and two 24GB RTX 3090 GPUs. Additionally, quantum circuit simulations are deployed on SJTU HPC "siyuan" No1, supporting larger-scale simulations. Detailed hyperparameters, including the learning rate and network parameters, can be found in the Supplementary Notes 6.

Data availability

All data needed to evaluate the conclusions are available from https://github.com/XiaoTailong/QDGP_GI_ImageTasks. The data include the bucket signals of biosamples including 'wing', 'butterfly' and 'plane' and the illumination patterns. The data also include the natural images testified in this work.

Code availability

The QDGP code is based on the implementation of DGP in https://github.com/XiaoTailong/QDGP_GI_ImageTasks. The code includes the implementation of GI and image restoration and manipulation.

Received: 8 December 2023; Accepted: 2 August 2024;
Published online: 15 August 2024

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Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 62071298), the fund of the State Key Laboratory of Advanced Optical Communication Systems and Networks, the Innovation Program for Quantum Science and Technology (Grant No. 2021ZD0300703), Shanghai Municipal Science and Technology Major Project (2019SHZDZX01) and SJTU-Lenovo Collaboration Project (202407SJTU01-LR019).

Author contributions

T.L.X. proposed the framework, developed the simulation, implemented the algorithms and elaborated on the quantum machine learning architecture design. X.L.Z. designed and conducted the ghost imaging experiments. X.L.Z., T.L.X., J.Z.H., and J.P.F. operated the classical machine learning analysis, and G.Z. conceived the research. All the authors discussed and contributed to the writing of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s42005-024-01765-9.

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Peer review information *Communications Physics* thanks the anonymous reviewers for their contribution to the peer review of this work. A peer review file is available.

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